

Ingestion

Indexes created first

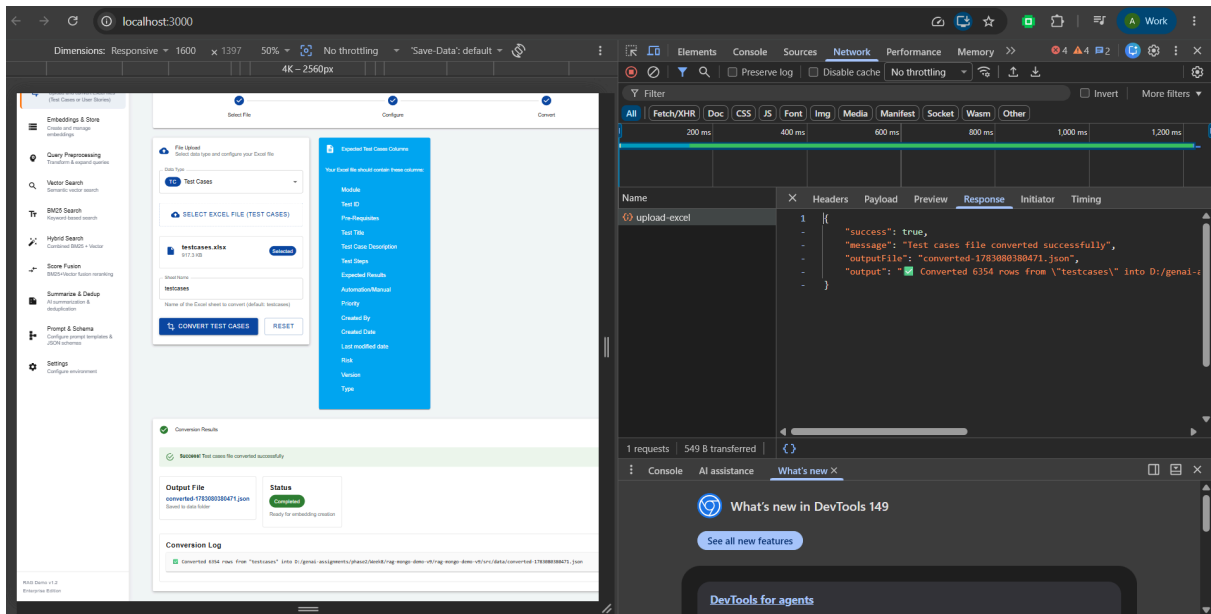
The first screenshot shows the MongoDB Atlas Data Explorer interface for the 'test_cases' collection. The left sidebar displays the 'Data Explorer' section with a tree view of clusters. The main panel shows the 'Indexes' tab for the 'test_cases' collection. A table lists the indexes created:

Name and Fields	Type	Status
testcase_bm25	SEARCH	READY
vector_index_arvind	VECTOR SEARCH	READY

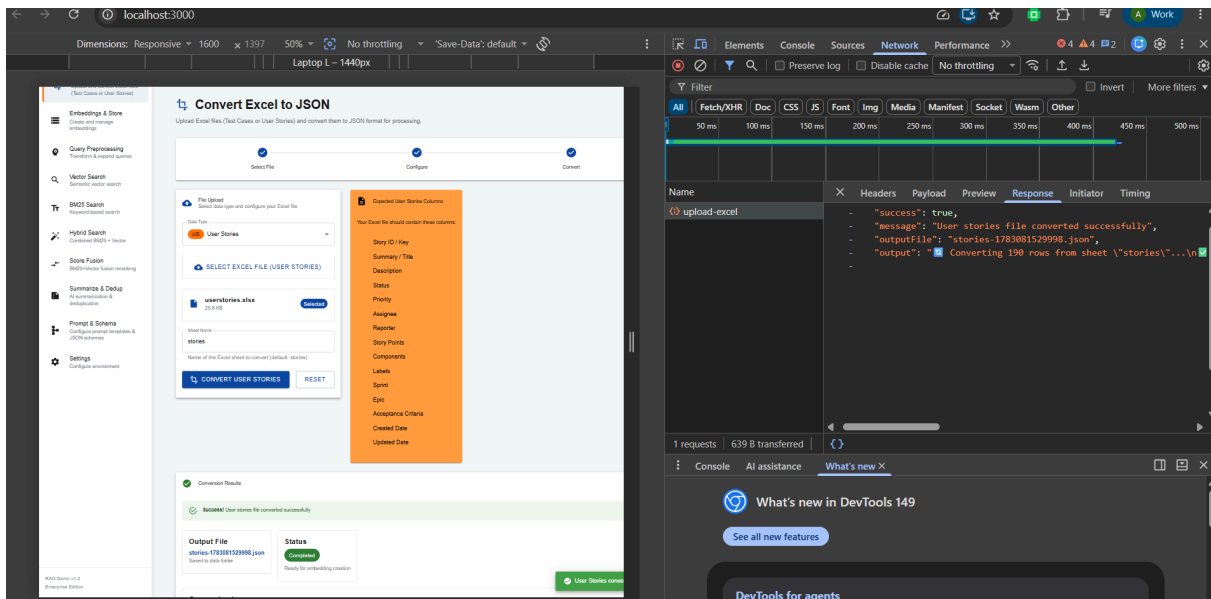
The second screenshot shows the same interface for the 'user_stories' collection. The 'Indexes' tab displays a single index:

Name and Fields	Type	Status
vector_index_userstories	VECTOR SEARCH	READY

Conversion of testcases excel into json using the application. Convert to Json option→First upload testcases and on clicking “Convert Testcases” button (done through api <http://localhost:3001/api/upload-excel> which can be seen in headers and response can be seen in)



Conversion of usrstories into json using the application. Convert to Json option→First upload usrstories excel and on clicking “Convert Userstories” button (done through api <http://localhost:3001/api/upload-excel> which can be seen in headers and response can be seen in)



All documents inserted in mongo db and Embeddings for testcases and userstories can be seen after Embeddings don on FE

```

PS D:\genai-assignments\phase2\Week8\rag-mongo-demo-v9\rag-mongo-demo-v9> npm run dev
[0]
[0] [user-stories] 🚀 [Batch 4/4] Processing 40 user stories with Mistral AI...
[0]
[0] [user-stories] ✅ [Batch 4/4] Success! Cost: $0.000641 | Tokens: 6413
[0]
[0] [user-stories] 📊 Progress: 190/190 (100.0%) | Rate: 26.1/sec | ETA: 0s | Cost: $0.003068
[0]
[0] [user-stories]
[0] 🎉 MISTRAL AI BATCH PROCESSING COMPLETE!
[0] 📊 Final Statistics:
[0]
[0] [user-stories] ⌚ Total Time: 8s
[0] ⚡ Processing Rate: 23.5 user stories/second
[0] 📄 Total User Stories: 190
[0] ✅ Successfully Processed: 190
[0] ❌ Failed: 0
[0] 📈 Success Rate: 100.0%
[0] 💰 Total Cost: $0.003068
[0]
[0] [user-stories] 🗣️ Total Tokens: 30,678
[0]
[0] [user-stories] 📊 Average Cost per User Story: $0.0001615
[0] 📊 Average Tokens per User Story: 161
[0] 💡 Cost Savings: ~90% cheaper than OpenAI embeddings
[0]
[0] ✅ Batch script completed successfully!

```

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS PLAYWRIGHT node
PS D:\genai-assignments\phase2\Week8\rag-mongo-demo-v9\rag-mongo-demo-v9> npm run dev
[0]
[0] [testcases]
[0] 🎉 MISTRAL AI BATCH PROCESSING COMPLETE!
[0] 📊 Final Statistics:
[0] ⌚ Total Time: 3m 14s
[0] ⚡ Processing Rate: 32.8 testcases/second
[0] 📄 Total Test Cases: 6354
[0] ✅ Successfully Processed: 6354
[0] ❌ Failed: 0
[0]
[0] [testcases] 📈 Success Rate: 100.0%
[0] 💰 Total Cost: $0.117741
[0]
[0] [testcases] 🗣️ Total Tokens: 11,77,411
[0] 📊 Average Cost per Test Case: $0.0001853
[0] 📊 Average Tokens per Test Case: 185
[0] 💡 Cost Savings: ~90% cheaper than OpenAI embeddings
[0]
[0] ✅ Batch script completed successfully!

```

cloud.mongodb.com/v2/6a2bd3301b2f148a67ddb85c?utm_source=DE&utm_medium=product#/explorer/6a2bd47bf4061564aa8f07c/db_testcases/test_cases/find

ORGANIZATION Arvind's Org - 2026-0... PROJECT Project 0

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

India86

- admin
- db_testcases
 - test_cases
- user_stories
- local

India86 > db_testcases > test_cases

Documents 6.4K Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#)

25 1 - 25 of 6354

```
{
  "_id": ObjectId("6a4f479831e45c386921a7bb"),
  "module": "Registration",
  "id": "TC_005",
  "prerequisites": "PRN/ERN Number",
  "title": "Conversion from PRN/ERN Reg to UHID",
  "description": "Verify Conversion from PRN/ERN Reg to UHID",
  "steps": "1. Launch the application\n2. Login to the application with valid creden...",
  "expectedResults": "Should get convert successfully from PRN/ERN Reg to UHID",
  "automationManual": "Automated",
  "priority": "P1",
  "createdBy": "Prabakaran",
  "createdAt": 45165,
  "lastModifiedDate": 45176,
  "linkedStories": "HC-888",
  "risk": "Low",
  "version": "v7.0",
  "type": "Integration"
}
```

Alert System Status: All Good

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cloud.mongodb.com/v2/6a2bd3301b2f148a67ddb85c?utm_source=DE&utm_medium=product#/explorer/6a2bd47bf4061564aa8f07c/db_testcases/user_stories/find

ORGANIZATION Arvind's Org - 2026-0... PROJECT Project 0

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

India86

- admin
- db_testcases
 - test_cases
- user_stories
- local

India86 > db_testcases > user_stories

Documents 190 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#)

25 1 - 25 of 190

```
{
  "issueLinks": Array (empty),
  "url": "",
  "sourceType": "excel",
  "importedAt": "2026-07-03T12:25:38.357Z",
  "originalRowIndex": 1,
  "embedding": Array (1824)
  0: -0.840802901953125
  1: 0.0287861167578125
  2: 0.851483154296875
  3: 0.025421142578125
  4: 0.0401611328125
  5: 0.036407470703125
  6: 0.01935388837899625
  7: -0.0283960664453125
  8: -0.006824493408203125
  9: 0.005767822265625
  10: -0.008080323486328125
  11: 0.00721923828125
}
```

Alert System Status: All Good

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Now Query Processing

localhost:3000

Navigation

- Convert to JSON
 - Upload and convert Excel files (Test Cases or User Stories)
- Embeddings & Store
 - Create and manage embeddings
- Query Preprocessing**
 - Transform & expand queries
- Vector Search
 - Semantic vector search
- BM25 Search
 - Keyword-based search
- Hybrid Search
 - Combined BM25 + Vector
- Score Fusion
 - BM25+Vector fusion reranking

RAG Pipeline

Home > Query Preprocessing Transform & expand queries

Query Preprocessing & Search

Transform your query through normalization, abbreviation expansion, and synonym generation before searching

Enter Your Query

Share Diagnostic Reports with Patients via WhatsApp

Quick Examples:

UHID patient login issue OTP not working TC_027 merge UHID records doctor appointment booking IP admission password reset OTP verification

ER patient registration BP monitoring

[PREPROCESS QUERY](#)

PIPELINE STEPS ABBREVIATIONS SYNONYM VARIATIONS

Navigation

Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization & deduplication

RAG Pipeline

Home > Query Preprocessing

Transform & expand queries

Query Transformation Pipeline

ORIGINAL QUERY

"Share Diagnostic Reports with Patients via WhatsApp"

1 NORMALIZATION

"share diagnostic reports with patients via whatsapp"

7 tokens

2 ABBREVIATION EXPANSION

0 expansions

"share diagnostic reports with patients via whatsapp"

3 SYNONYM EXPANSION

1 variations

Multiple query variations generated for comprehensive search coverage

1. "share diagnostic reports with patients via whatsapp"

Processing Metadata

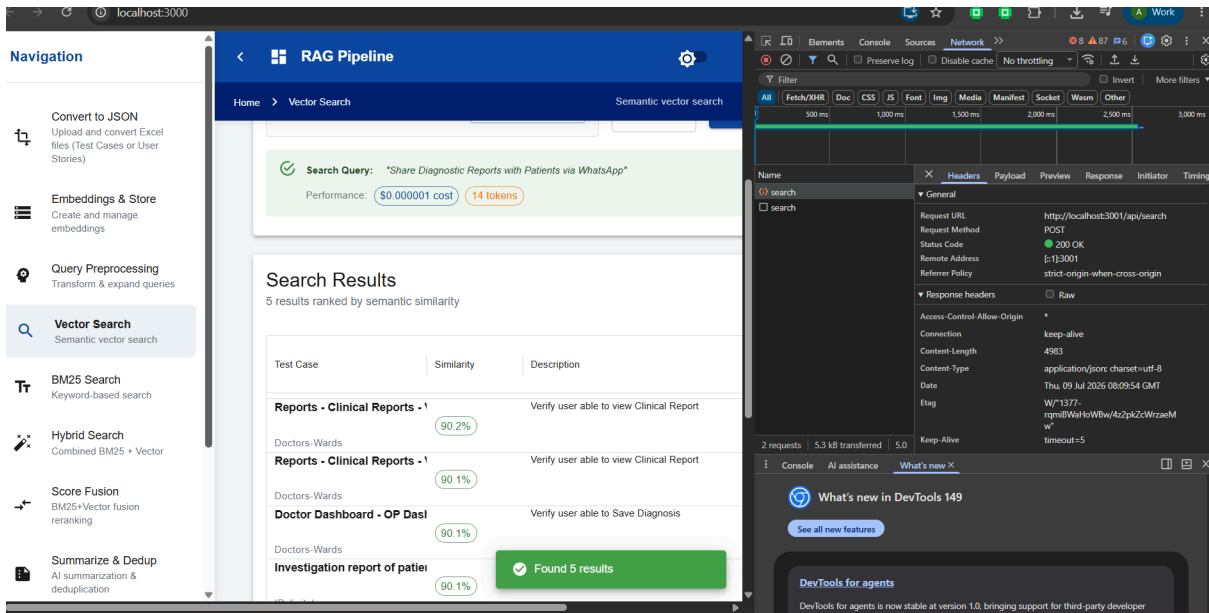
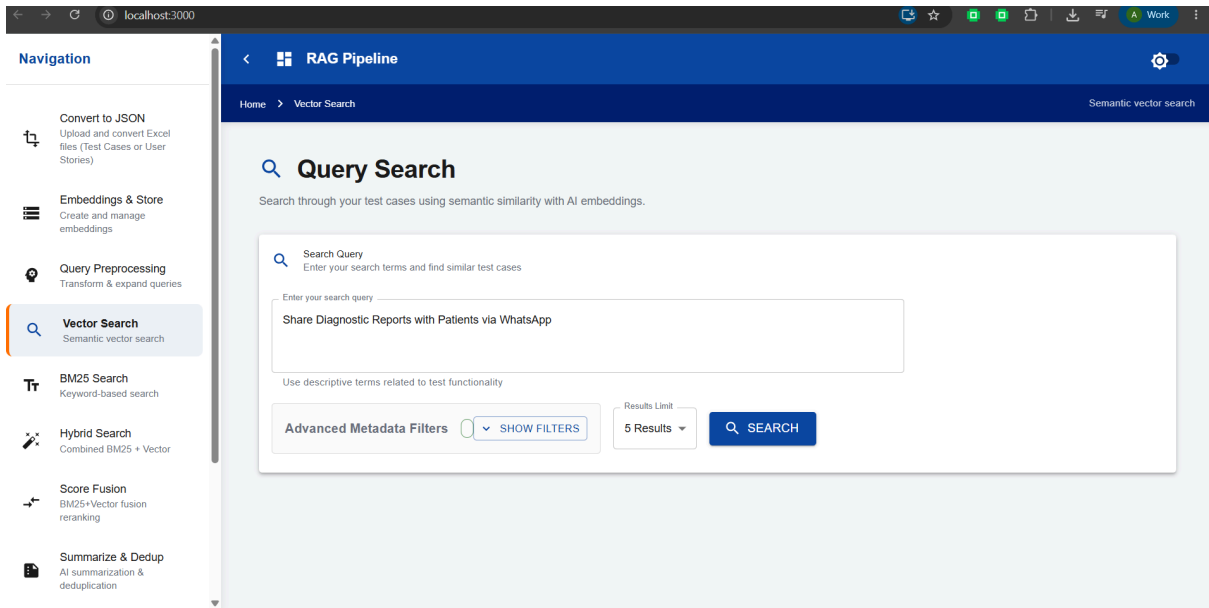
Processing Time

Token Count

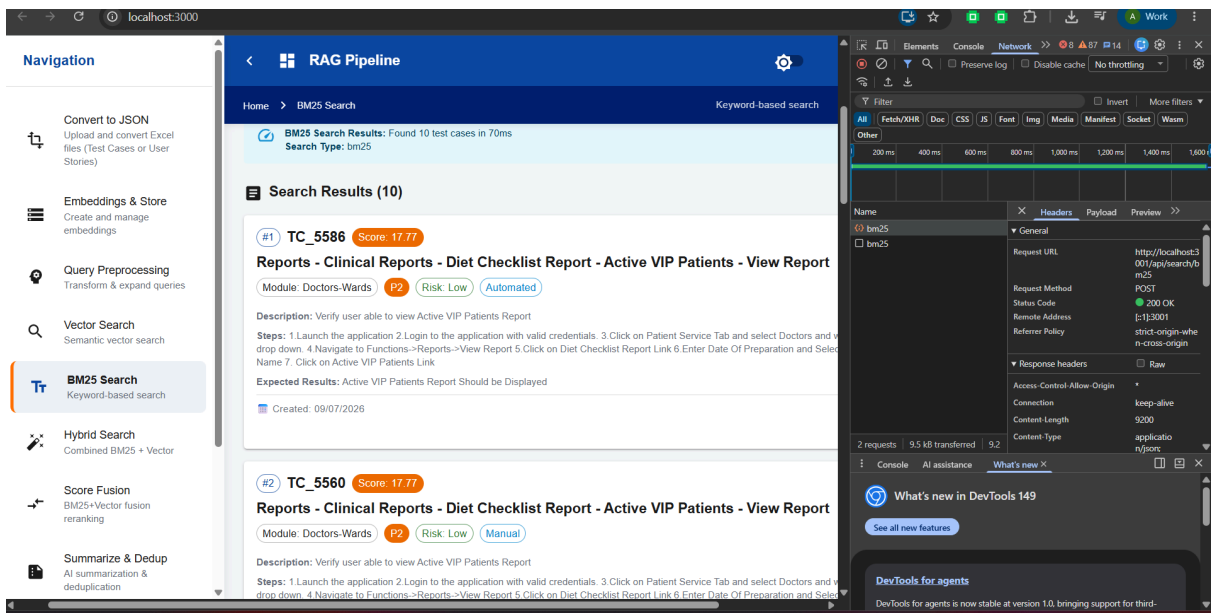
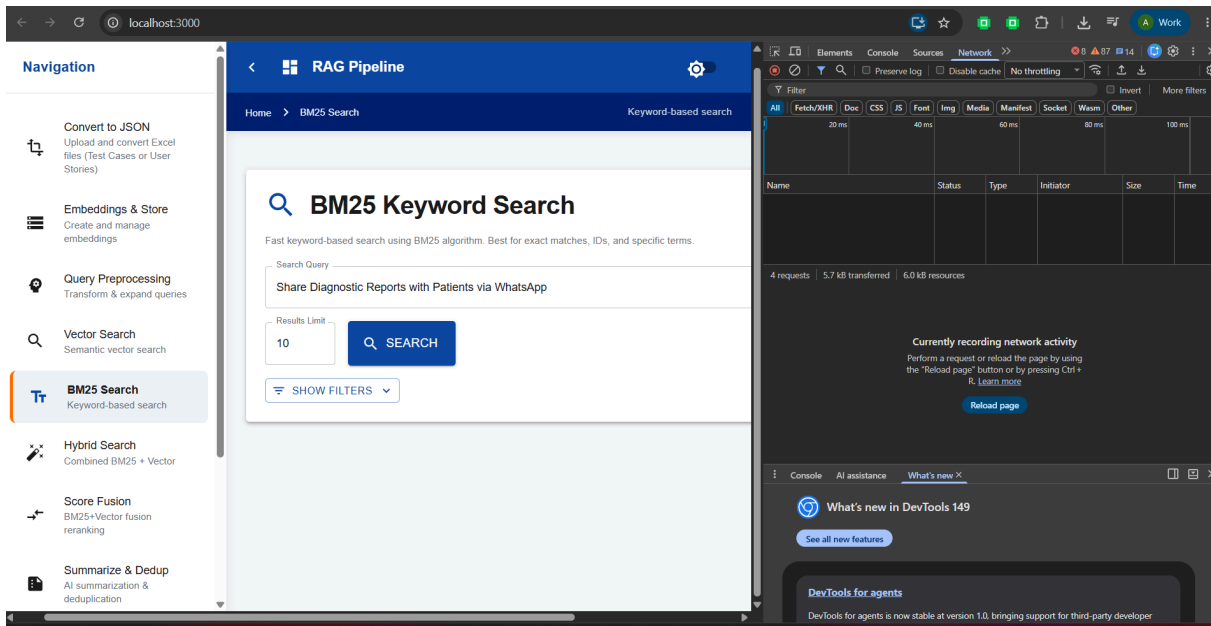
Abbreviations

Synonyms

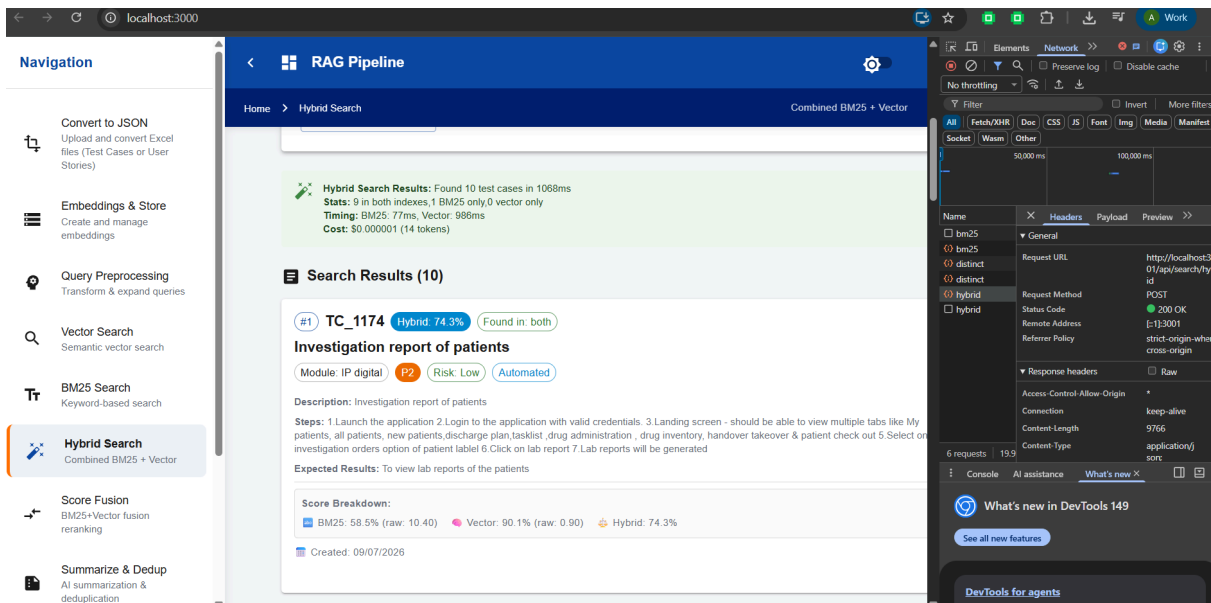
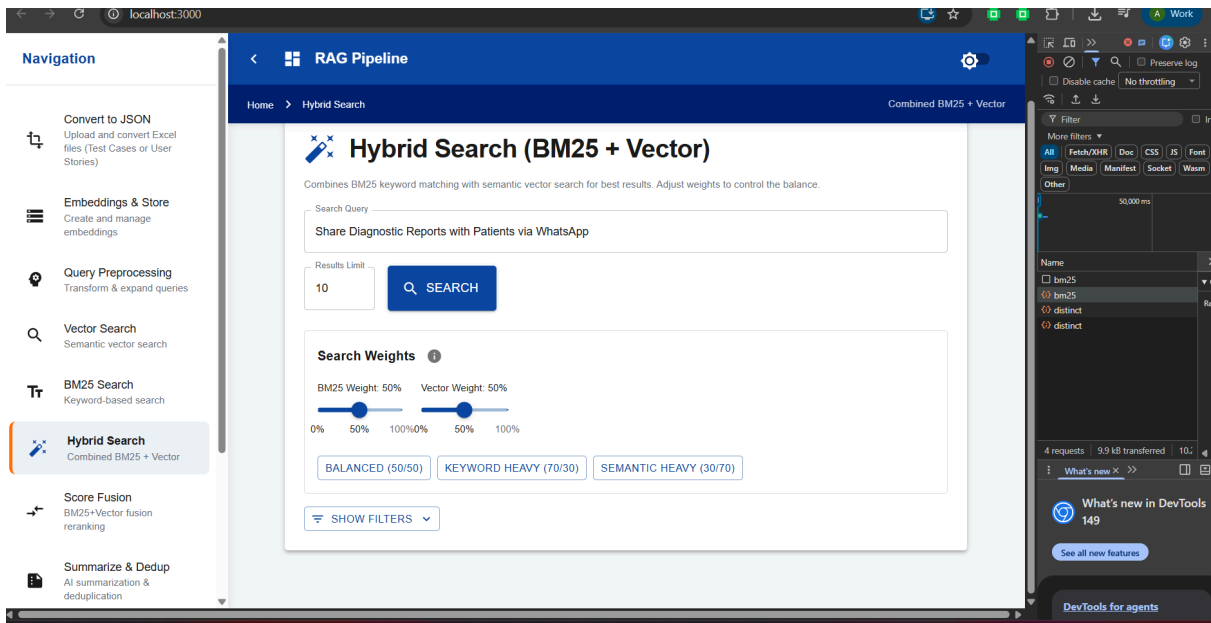
Vector Search



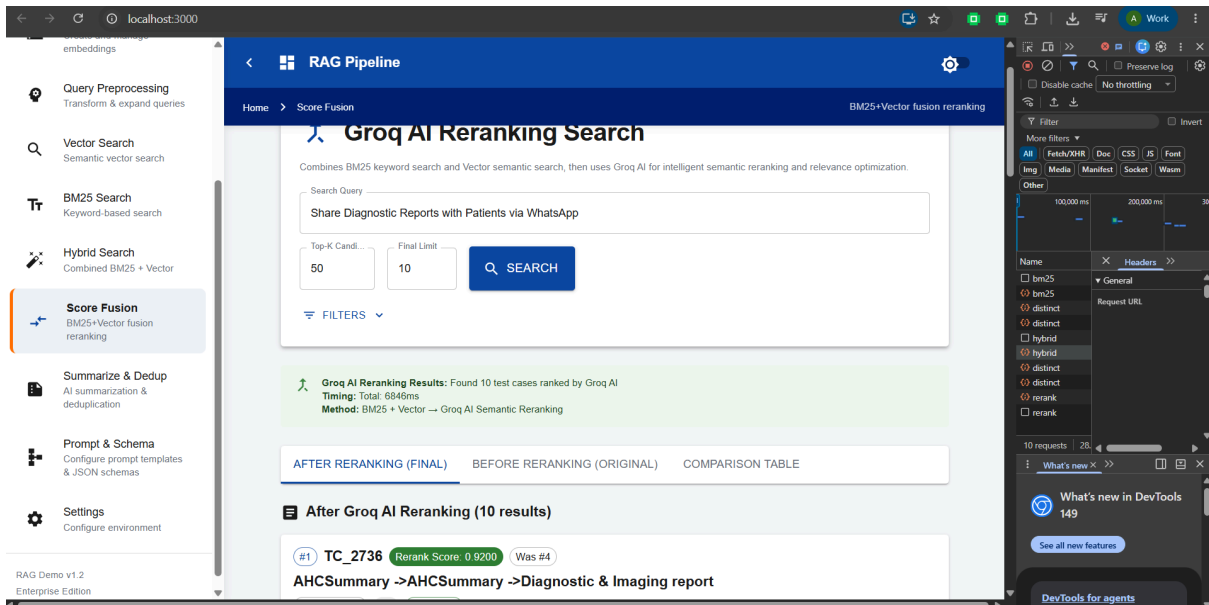
BM 25 Search



Hybrid Search(Vector+BM25)

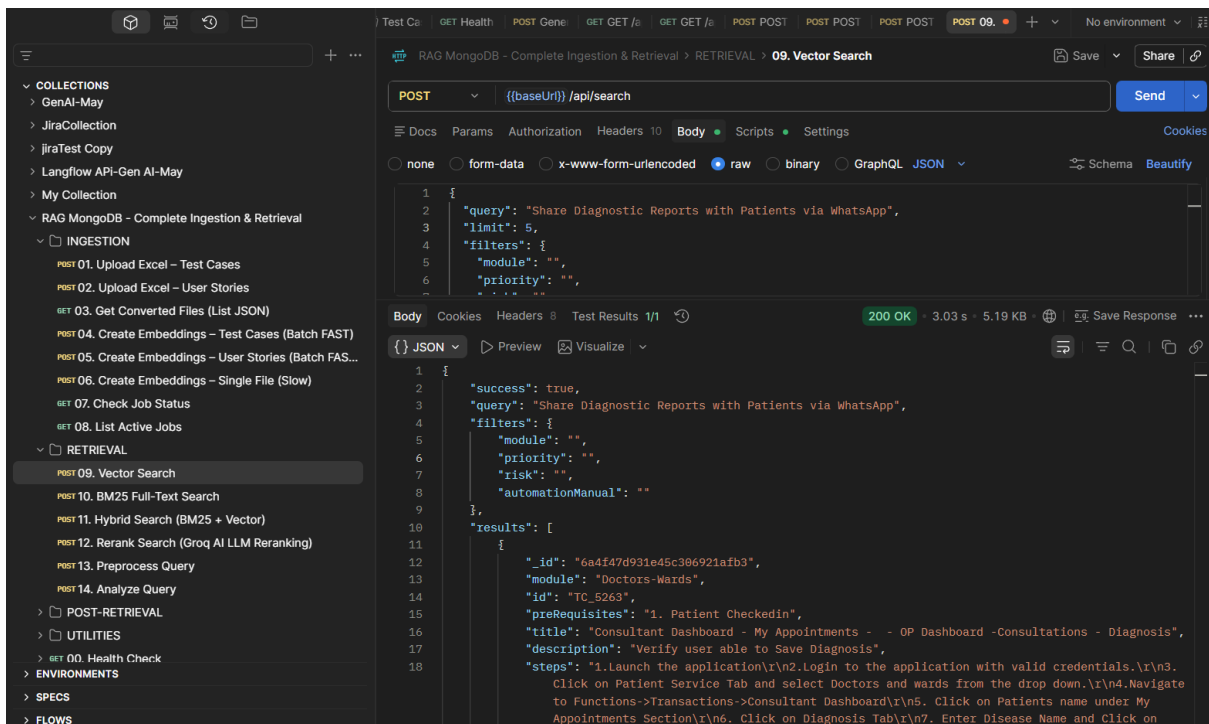


Reranking



Ingestion and retrieval through postman collections

Vector search



BM25 Search

REST client interface showing a POST request to `/api/search/bm25`. The request body is a JSON object:

```
{
  "query": "Share Diagnostic Reports with Patients via WhatsApp",
  "limit": 3,
  "filters": {
    "module": "",
    "priority": "",
    "risk": ""
  }
}
```

The response is a 200 OK status with a JSON body:

```
{
  "results": [
    {
      "_id": "6a4f47f631e45c386921b321",
      "module": "Doctors-Wards",
      "id": "TC_5586",
      "title": "Reports - Clinical Reports - Diet Checklist Report - \r\nActive VIP Patients - View Report",
      "description": "Verify user able to view Active VIP Patients Report",
      "steps": "1.Launch the application\r\n2.Login to the application with valid credentials.\r\n3. Click on Patient Service Tab and select Doctors and wards from the drop down.\r\n4.Navigate to Functions->Reports->View Report\r\n5.Click on Diet Checklist Report Link\r\n6.Enter Date Of Preparation and Select Service Name\r\n7. Click on Active VIP Patients Link",
      "expectedResults": "\r\nActive VIP Patients Report Should be Displayed",
      "automationManual": "Automated",
      "priority": "P2",
      "risk": "Low"
    }
  ]
}
```

Hybrid search

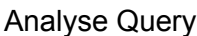
REST client interface showing a POST request to `/api/search/hybrid`. The request body is a JSON object:

```
{
  "query": "Share Diagnostic Reports with Patients via WhatsApp",
  "limit": 1,
  "bm25Weight": 0.4,
  "vectorWeight": 0.6,
  "filters": {
    "module": "",
    "priority": ""
  }
}
```

The response is a 200 OK status with a JSON body:

```
{
  "weights": {
    "bm25": 0.4,
    "vector": 0.6
  },
  "results": [
    {
      "_id": "6a4f47d931e45c386921afb3",
      "module": "Doctors-Wards",
      "id": "TC_5263",
      "title": "Consultant Dashboard - My Appointments - - OP Dashboard -Consultations - Diagnosis",
      "description": "Verify user able to Save Diagnosis",
      "steps": "1.Launch the application\r\n2.Login to the application with valid credentials.\r\n3. Click on Patient Service Tab and select Doctors and wards from the drop down.\r\n4.Navigate to Functions->Transactions->Consultant Dashboard\r\n5. Click on Patients name under My Appointments Section\r\n6. Click on Diagnosis Tab\r\n7. Enter Disease Name and Click on Search\r\n8.Select disease and click on Add\r\n9. Click on Save as Draft",
      "expectedResults": "Draft Saved Successfully Message should be displayed",
      "automationManual": "Automated",
      "priority": "P2"
    }
  ]
}
```

Reranking



COLLECTIONS

GenAI-May

JiraCollection

jiraTest Copy

Langflow API-Gen AI-May

My Collection

RAG MongoDB - Complete Ingestion & Retrieval

INGESTION

POST 01. Upload Excel - Test Cases

POST 02. Upload Excel - User Stories

GET 03. Get Converted Files (List JSON)

POST 04. Create Embeddings - Test Cases (Batch FAST)

POST 05. Create Embeddings - User Stories (Batch FAS...

POST 06. Create Embeddings - Single File (Slow)

GET 07. Check Job Status

GET 08. List Active Jobs

RETRIEVAL

POST 09. Vector Search

POST 10. BM25 Full-Text Search

POST 11. Hybrid Search (BM25 + Vector)

POST 12. Rerank Search (Groq AI LLM Reranking)

POST 13. Preprocess Query

POST 14. Analyze Query

POST-RETRIEVAL

UTILITIES

GET 00. Health Check

ENVIRONMENTS

SPECS

FLOWS

RAG MongoDB - Complete Ingestion & Retrieval > RETRIEVAL > 14. Analyze Query

POST {{baseUrl}} /api/search/analyze

Send

Docs Params Authorization Headers 10 Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "query": "doctor appointment booking ip admission"
3 }
```

Body Cookies Headers 8 Test Results 200 OK 13 ms 982 B Save Response

JSON Preview Visualize

```
14 ],
15   "hasSynonymOpportunities": true,
16   "synonymOpportunities": [
17     {
18       "term": "doctor",
19       "synonyms": [
20         "physician",
21         "consultant",
22         "medical officer",
23         "provider",
24         "clinician"
25       ],
26       "position": 0
27     },
28     {
29       "term": "appointment",
30       "synonyms": [
31         "booking",
32         "scheduling",
33         "meeting",
34         "consultation"
35       ],
36       "position": 1
37     }
38   ]
39 }
```

Connect Git Console Terminal 0 0 Import Complete